



F-17 Humanitarian Hummingbird

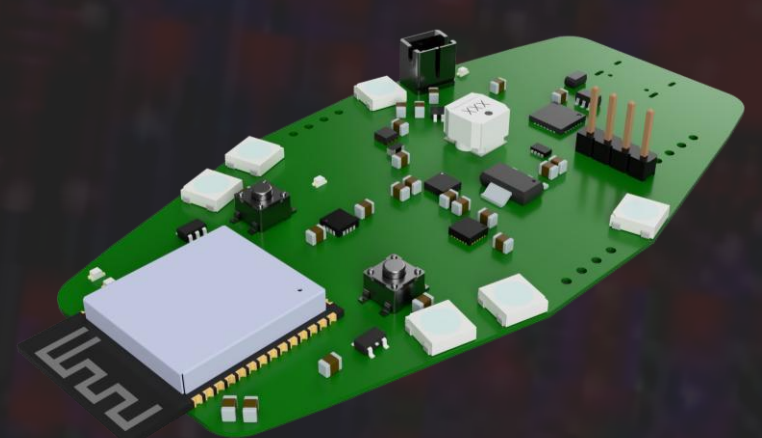
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Problem Statement

No sub-250g, modular, affordable microdrone exists on the market. Addresses that gap through an adaptable platform for surveying and UAV applications.

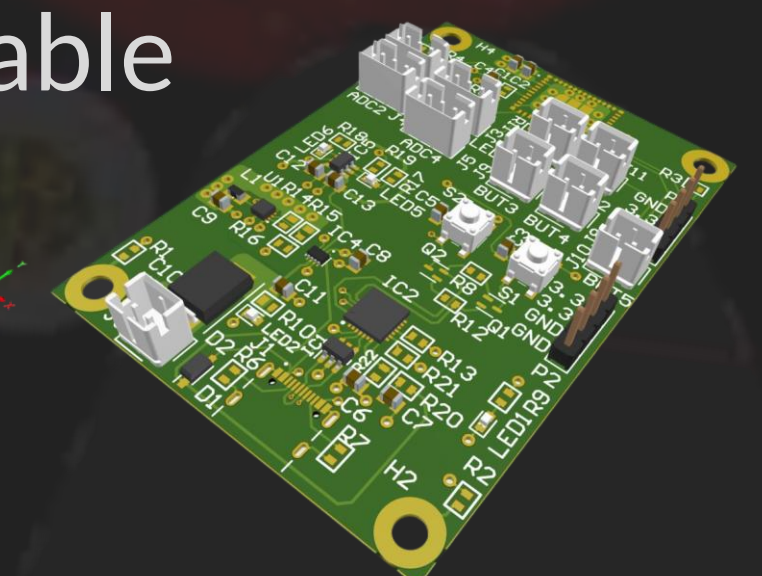
Flight Controls

Wirelessly communicated flight controls drive a PID loop utilizing a Mahony filter to achieve flight attitude targets. Specifies motor speeds using 50Hz PWM to ESCs.



Communications

The controller relays two bi-axial gimbal controls, four buttons, and one arming switch to the flight control board via ESP-NOW. Implements fault disable safety logic and allows a range of over 20m.



Testing & Results

7 custom PCBs. Over 3000 lines of code. Achieves stable control for ~5min flight times. Set up for modular development and autonomous control expansions.

Mechanical Design

Thrust to weight ratio up to 2.32 and light weight construction means incredible maneuverability.



Brushless DC Motor Driver

Centered around a STM32f4 running bare-metal C firmware to control four brushless DC motors via 12 half bridges.

Sensorless interrupt driven trapezoidal six-step commutation allows for efficiencies over 50% better than off-the-shelf alternatives.

